

Retrieval Augmented Generation

RAG

Enhancing Large Language Models with External Knowledge

What is RAG?

Retrieval Augmented Generation (RAG) is a technique that improves Large Language Models by retrieving relevant information from external data sources before generating a response. Instead of relying solely on pre-trained knowledge, RAG dynamically fetches context to ground the model's output in accurate, up-to-date information.

Key Benefits

Reduces Hallucinations	Grounds responses in retrieved facts, minimizing fabricated content.
Uses Private Company Data	Connects LLMs to internal knowledge bases and proprietary documents.
Provides Up-to-Date Information	Retrieves current data beyond the model's training cutoff.
Improves Factual Accuracy	Verifiable sources back every generated response.

RAG Workflow

1	User asks a question The user submits a natural language query to the system.
2	Convert question to embedding The query is transformed into a vector representation using an embedding model.
3	Search vector database The embedding is used to perform a similarity search across the vector store.
4	Retrieve relevant chunks The most semantically similar document chunks are fetched.
5	Send chunks to LLM Retrieved context is combined with the original query and passed to the language model.
6	Generate final answer The LLM produces a grounded, accurate response using the provided context.

Popular Vector Databases

Qdrant	Open-source, high-performance vector DB with filtering support.
Pinecone	Fully managed, cloud-native vector search service.
Weaviate	Open-source with built-in ML model integrations.
Milvus	Scalable, open-source vector DB designed for AI applications.